FTEC5660 Reproducibility Report: TradingAgents

Multi-Agents LLM Financial Trading Framework

Course: FTEC5660 Agentic AI for Business and FinTech

Assignment: Reproducibility Work

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1. Project Summary

1.1 Paper Overview

TradingAgents (Xiao et al., 2024) is a multi-agent LLM framework for financial trading that simulates a real-world trading firm. The system employs multiple specialized LLM-based agents:

• **4 Analysts**: Market Analyst (technical indicators), Sentiment Analyst (StockTwits/Reddit/news), News Analyst (macro news), Fundamentals Analyst (SEC filings)

• **2 Researchers**: Bull Researcher and Bear Researcher debate based on analyst reports

• **1 Research Manager**: Synthesizes the debate into an investment recommendation

• **1 Trader**: Converts the recommendation into a trade action (Buy/Sell/Hold)

• **3 Risk Debaters**: Aggressive, Conservative, and Neutral analysts debate risk

• **1 Portfolio Manager**: Makes the final decision

The paper reports (Table 1) that TradingAgents achieved 26.62% cumulative return on AAPL, 24.36% on GOOGL, and 23.21% on AMZN over Jan-Mar 2024, significantly outperforming Buy & Hold and technical indicator baselines (MACD, KDJ&RSI;, ZMR, SMA).

1.2 What We Tried to Reproduce

We targeted reproducing the core multi-agent decision pipeline: given a stock ticker and date, the system should produce a trading decision (Buy/Overweight/Hold/Underweight/Sell) through its full agent chain.

Specific reproduction target: Run TradingAgents for NVDA and AAPL on 2 dates each in January 2024, verify the pipeline executes end-to-end, and record decisions.

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2. Setup Notes

2.1 Environment

Component	Specification
OS	macOS (Darwin)
Python	3.13.12
Virtual Env	venv (isolated)
LLM Provider	DeepSeek (replacing OpenAI)
Deep Think Model	deepseek-reasoner (replacing o1-preview)
Quick Think Model	deepseek-chat (replacing gpt-4o)
Temperature	0.0 (for reproducibility)
Debate Rounds	1 (bull-bear)
Risk Debate Rounds	1 (aggressive-conservative-neutral)

2.2 Data Sources

Source	Status	Notes
SEC EDGAR	■ Working	Free, fundamental data (as-filed) — only source with real data
Yahoo Finance	■ HTTP 429	OHLCV, indicators, stock news, global news — all rate limited
StockTwits	■ 403 Forbidden	Blocked access — no retail sentiment
Reddit RSS	■ Timeout	Connection issues — no Reddit sentiment
FRED	■ Not configured	FRED_API_KEY not set — no macro indicators (CPI, PCE, etc.)
Polymarket	■■ Withheld	Live odds only; no historical vintage → withheld to avoid look-ahead bias

> **Important**: Despite all sentiment/news/market data sources failing, the Sentiment Analyst and News Analyst still *generated reports*. These are "data-unavailable" placeholder texts that explicitly decline to fabricate data. The JSON metadata `has_sentiment_data: true` / `has_news_data: true` reflects that an LLM produced text, **not** that actual data was retrieved. See Section 4.3 for full explanation.

2.3 Model Swap Justification (Edge Case A)

The original paper uses OpenAI's o1-preview and gpt-4o. We replaced these with DeepSeek's deepseek-reasoner and deepseek-chat respectively, as explicitly allowed by the assignment (Section 4, Edge Case A).

Documentation of changes:

• Model name: `o1-preview` → `deepseek-reasoner`, `gpt-4o` → `deepseek-chat`

• Provider: `openai` → `deepseek`

• Base URL: Automatically configured by TradingAgents (DeepSeek API is OpenAI-compatible)

• Decoding params: temperature=0.0 (same for both)

• No prompt/template changes needed

• **Results are not directly comparable to the paper because the underlying LLM differs**

2.4 API Keys

• DeepSeek API key configured in `.env` (not committed to git)

• No other paid API keys required

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3. Reproduction Target & Metric

3.1 Target

Target claim: "TradingAgents produces multi-step trading decisions through its agent pipeline, with analysts providing reports, researchers debating, and a portfolio manager making a final call."

Metric: Final decision rating (Buy/Overweight/Hold/Underweight/Sell) and pipeline execution success (all agents producing reports).

3.2 Evaluation

We evaluate:

Whether the full pipeline executes end-to-end (all 11+ agent steps complete)

Whether each analyst produces a report

Whether a final decision is produced

Whether decisions are directionally reasonable (e.g., NVDA bullish in Jan 2024 during AI boom)

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4. Results: Our Numbers vs Reported

4.1 Baseline Reproduction Results

Ticker	Date	Decision	Has Fundamentals	Has Market Data	Has Sentiment Data	Has News Data	Time (s)
NVDA	2024-01-10	**Overweight**	■	■ (HTTP 429)	■■ placeholder†	■■ placeholder†	165.6
AAPL	2024-01-11	**Hold**	■	■ (HTTP 429)	■■ placeholder†	■■ placeholder†	197.1

Ticker	Date	Decision	Has Fundamentals	Has Market Data	Has Sentiment Data	Has News Data	Time (s)
NVDA	2024-01-25	**Overweight**	■	■ (HTTP 429)	■■ placeholder†	■■ placeholder†	163.0
AAPL	2024-01-25	**Hold**	■	■ (HTTP 429)	■■ placeholder†	■■ placeholder†	169.8

† **Placeholder note**: The `has_sentiment_data` and `has_news_data` fields in the JSON are marked `true` because the Sentiment Analyst and News Analyst each *produced a report* (an LLM-generated text output). However, **all underlying data sources returned UNAVAILABLE** — StockTwits (403), Reddit (timeout), Yahoo Finance news (HTTP 429), and FRED macro (key not set). The generated reports are "data-unavailable" placeholders that explicitly decline to fabricate data. No actual sentiment signals, news headlines, or macro indicators were retrieved. See Section 4.3, Observation 3 for details.

4.2 Comparison with Paper

Aspect	Paper	Our Reproduction
LLM	OpenAI o1-preview + gpt-4o	DeepSeek reasoner + chat
Data source	FinnHub (paid)	Yahoo Finance (free, rate-limited) + SEC EDGAR
Stocks tested	AAPL, GOOGL, AMZN	NVDA, AAPL
Period	Jan-Mar 2024 (daily)	Jan 2024 (4 dates)
Decision for NVDA	N/A (not in paper)	Overweight (consistent across dates)
Decision for AAPL	Bullish (26.62% CR)	Hold (more conservative)
Full pipeline execution	■	■
Multi-agent debate	■	■

Note on time values: All elapsed times are exact values from `reproduction_results/reproduction_summary.json` and `ablation_summary.json` (e.g., 165.6s, not 166s). Times are wall-clock measurements from script start to final decision; they include LLM API latency and data-source retry waits.

4.3 Key Observations

Pipeline integrity: The full multi-agent pipeline executes correctly with DeepSeek — all 4 analysts, 2 researchers, research manager, trader, 3 risk debaters, and portfolio manager complete their roles.

Fundamental analysis quality: SEC EDGAR data was retrieved successfully. The Fundamentals Analyst produced detailed, accurate reports (e.g., NVDA Q3 FY2024: revenue $18.12B, +206% YoY, gross margin 74.0%).

Sentiment and News data are placeholders, not real data: The JSON metadata fields `has_sentiment_data` and `has_news_data` are `true` because the Sentiment Analyst and News Analyst each *generated a text report*. However, **every underlying data source returned UNAVAILABLE**:

• **Sentiment Analyst**: StockTwits (403 Forbidden), Reddit RSS (timeout) → no retail sentiment signals

• **News Analyst**: Yahoo Finance news (HTTP 429), FRED macro (API key not set) → no news headlines or macro indicators

The generated reports are "data-unavailable" meta-reports that explicitly state "I will not fabricate data" and assign a default Neutral score (5.0/10) with Low confidence. **No actual sentiment signals, news headlines, or macro indicators were retrieved.** The `has_sentiment_data: true` / `has_news_data: true` flags in the JSON reflect "an LLM produced text," not "data was available." This is a known limitation of our reproduction script's metadata labeling — the flags should be interpreted as "agent ran" rather than "data was retrieved."

Market data gap: Yahoo Finance rate limiting (HTTP 429) prevented OHLCV, technical indicators, and price data from being retrieved. The Market Analyst's report is also a placeholder stating "DATA_UNAVAILABLE."

Decision rationale: Decisions were driven **solely by fundamental analysis** (SEC EDGAR data). Despite missing market, sentiment, and news data, the system made reasonable decisions:

• NVDA → Overweight: driven by +206% YoY revenue growth and margin expansion

• AAPL → Hold: more conservative given slower growth and valuation concerns

Paper's NVDA comparison: NVDA was not in the paper's Table 1, but our Overweight rating aligns with NVDA's actual ~80% price increase in Q1 2024.

AAPL divergence: The paper reports strong bullish performance on AAPL (26.62% CR), but our system gave "Hold." This divergence is likely due to:

• Different LLM (DeepSeek vs OpenAI)

• Missing market data (no price/technical indicators)

• Different time period (single dates vs daily compounding)

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5. Modification & Results

5.1 Modification: Ablation — Remove Risk Debate

What changed: Set `max_risk_discuss_rounds=0`, which skips the Aggressive/Conservative/Neutral risk analysts' debate entirely. The pipeline goes directly from Trader → Portfolio Manager.

Why: The paper claims risk management (Section 3.4) as a key component. We test whether removing it changes decisions.

Isolation: Only one parameter changed (`max_risk_discuss_rounds: 1 → 0`); all other settings identical.

5.2 Ablation Results

Ticker	Date	Baseline Decision	Ablation Decision	Changed?	Baseline Time (s)	Ablation Time (s)
NVDA	2024-01-10	Overweight	Overweight	■ No	165.6	169.9
AAPL	2024-01-11	Hold	Hold	■ No	197.1	145.7

5.3 Modification Impact Analysis

Finding: The risk debate has no effect on the final decision rating.

Both NVDA and AAPL received identical ratings with and without the risk debate component. This suggests that in the current architecture, the Portfolio Manager largely follows the Research Manager's recommendation regardless of the risk analysts' input.

However, examining the ablation run's internal state reveals important nuances:

Risk debate still partially executes: Even with `max_risk_discuss_rounds=0`, the Aggressive Analyst still produces one statement (visible in `risk_debate_state.aggressive_history`). The conditional logic (`should_continue_risk_analysis`) checks if `count >= 3 * max_risk_discuss_rounds` (= 0), which is immediately true, so it routes to Portfolio Manager after the first speaker. The Aggressive Analyst gets exactly one turn but the Conservative and Neutral analysts get zero turns.

Reasoning depth differs: In the baseline run, the Portfolio Manager's final decision references "the conservative is right that..." and "the aggressive's bounded downside does not hold," indicating it incorporated all three risk perspectives. In the ablation run, the Portfolio Manager's reasoning references only the Aggressive Analyst's points.

Speed improvement for AAPL: The ablation run was ~26.0% faster for AAPL (145.7s vs 197.1s), as expected from skipping most of the risk debate.

Decision stability: The identical decisions suggest the system's core signal comes from the fundamental analysis + bull/bear debate, not the risk debate. The risk debate appears to be a "sanity check" layer that, in these cases, did not override the fundamental signal.

Implications for the paper's architecture claim: The paper positions risk management as a key component (Section 3.4), but our ablation suggests it may be ceremonial rather than decision-changing for these test cases. This could be because:

• The missing market data already makes the system conservative (no Buy → only Overweight)

• DeepSeek may be less influenced by additional debate context than OpenAI's models

• The test cases (NVDA strong fundamentals, AAPL stable) had clear directional signals

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6. Debug Diary

Blocker 1: Yahoo Finance Rate Limiting

• **Problem**: HTTP 429 "Too Many Requests" on all Yahoo Finance endpoints (stock data, indicators, news, global news)

• **Impact**: Market Analyst, Sentiment Analyst, and News Analyst received no data

• **Resolution**: Could not resolve within session. SEC EDGAR (fundamentals) worked reliably. The system's error handling correctly propagated "DATA_UNAVAILABLE" to agents rather than crashing.

• **Workaround**: Relied on fundamentals-only analysis for decision-making

Blocker 2: StockTwits 403 Forbidden

• **Problem**: StockTwits API returns HTTP 403

• **Impact**: Sentiment Analyst gets no retail sentiment data

• **Resolution**: No resolution; StockTwits appears to have changed access policy. System handles gracefully.

Blocker 3: Reddit RSS Timeout

• **Problem**: Reddit RSS feed requests timeout

• **Impact**: No Reddit sentiment data

• **Resolution**: System handles gracefully; no crash

Blocker 4: FRED API Key Not Set

• **Problem**: FRED_API_KEY not configured

• **Impact**: No macro economic data (CPI, PCE, unemployment, Fed funds rate)

• **Resolution**: Free key available at https://fred.stlouisfed.org; not critical for reproduction as fundamentals worked

Blocker 5: DeepSeek Model Warning

• **Problem**: RuntimeWarning "Model 'deepseek-reasoner' is not in the known model list for provider 'deepseek'"

• **Impact**: None — warning only, the model works correctly

• **Resolution**: Ignored; the framework's model list is outdated

Blocker 6: pip Installation Issues

• **Problem**: First `pip install -e .` appeared to succeed but `import tradingagents` failed; then EEXIST error on rebuild

• **Resolution**: `pip cache purge` + reinstall fixed the issue

Non-Blocker: Research Manager Structured Output

• **Problem**: "structured output returned no parsed result; retrying once as free text"

• **Impact**: The Research Manager's structured output failed once but the system retried successfully

• **Resolution**: Self-healing; no action needed

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7. Conclusions

7.1 What Is Reproducible

Multi-agent pipeline execution: ■ Fully reproducible. The complete agent chain (analysts → researchers → debate → trader → risk debate → portfolio manager) runs end-to-end with a different LLM provider (DeepSeek).

Fundamental analysis quality: ■ SEC EDGAR data retrieval and the Fundamentals Analyst's report are high quality and accurate. NVDA's Q3 FY2024 financials were correctly extracted and analyzed.

Decision production: ■ The system consistently produces a clear 5-tier rating (Buy/Overweight/Hold/Underweight/Sell).

Robustness to data failures: ■ The system gracefully handles data source failures, reporting "DATA_UNAVAILABLE" rather than fabricating data or crashing.

7.2 What Is Not Reproducible

Full backtest comparison: ■ Yahoo Finance rate limiting prevented us from running the full daily backtest needed to compute CR%, ARR%, SR, MDD% as in Table 1.

Market technical analysis: ■ No OHLCV, RSI, MACD, or moving average data was available due to rate limiting.

Sentiment analysis: ■ StockTwits (403) and Reddit (timeout) both failed. The Sentiment Analyst generated a placeholder report with a default Neutral score (5.0/10, Low confidence) but **no actual sentiment data was retrieved**. The JSON flag `has_sentiment_data: true` is misleading — it means "agent produced text," not "data was available."

News analysis: ■ Yahoo Finance news (HTTP 429) and FRED macro (key not set) both failed. The News Analyst generated a placeholder report stating "zero usable observations" but **no actual news or macro data was retrieved**. The JSON flag `has_news_data: true` is similarly misleading.

Exact numerical comparison: ■ Since we used DeepSeek instead of OpenAI, and many data sources failed, exact numerical comparison with the paper's Table 1 is not possible.

7.3 Why

External API limitations: Yahoo Finance's free tier has aggressive rate limits that made systematic data retrieval impossible during our testing window. The paper used FinnHub (paid) which likely had higher quotas.

Model differences: DeepSeek and OpenAI models have different reasoning patterns. DeepSeek produced more conservative decisions (e.g., "Hold" for AAPL vs the paper's bullish results), which could reflect genuine model differences or the impact of missing market data.

Non-determinism: Even at temperature=0, LLM outputs vary across runs. The paper itself acknowledges this limitation.

Temporal data drift: News, sentiment, and social media data are inherently non-reproducible for historical dates — the paper acknowledges this as a fundamental challenge.

7.4 Recommendations for Future Users

Use Alpha Vantage as a fallback data source: Configure `ALPHA_VANTAGE_API_KEY` to avoid Yahoo Finance rate limits

Set FRED_API_KEY: Free and enables macro economic data

Budget for API costs: Each run requires ~11 LLM calls + 20+ tool calls; DeepSeek is ~10x cheaper than OpenAI

Run multiple trials: Due to LLM non-determinism, run 3-5 trials per condition and report mean/variance

Pin data versions: For true reproducibility, cache Yahoo Finance responses locally

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References

Xiao, Y., Sun, E., Luo, D., & Wang, W. (2024). TradingAgents: Multi-Agents LLM Financial Trading Framework. arXiv:2412.20138.

TradingAgents GitHub: https://github.com/TauricResearch/TradingAgents

DeepSeek API: https://platform.deepseek.com/

SEC EDGAR: https://www.sec.gov/edgar/

Yahoo Finance API: https://finance.yahoo.com/

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